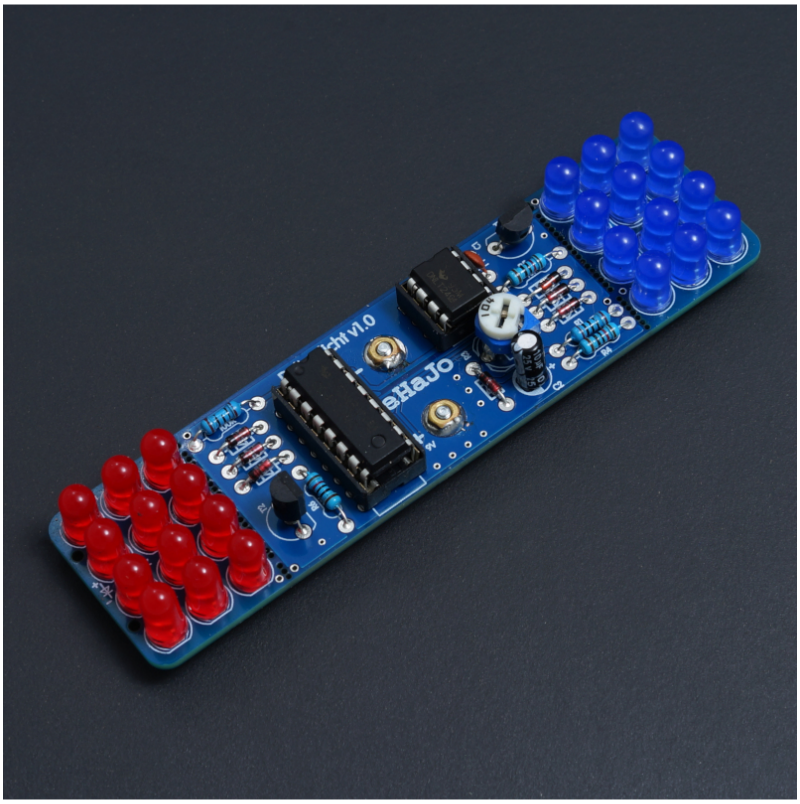
eHaJo
Elektronische Bausätze

Solderexcercise police light

Instructions



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Contents

1 Risk Assessment and Disposal

1.1 Risk Assessment

This kit is not suitable for children under 10 years of age. Assembly and use should always be supervised by an adult. Please note the following risks:

- Small parts can be swallowed. Keep the kit out of reach of small children.
- Tools such as tweezers or side cutters can cause injuries.
- The soldering iron becomes very hot and can cause burns.
- Use a fireproof surface to protect work areas.
- Flux can splatter during soldering. Wear protective goggles.
- Solder fumes can be harmful to health. Ensure proper ventilation.
- Incorrect assembly can result in malfunction. Follow the instructions.
- Mechanical damage can occur from improper handling.
- Components can be damaged by electrostatic discharge (ESD).
- Improper assembly can lead to short circuits.
- Wires on components are sharp and can cause puncture injuries.
- Use pliers to handle wires safely.
- Wires can break and fly off. Protect your eyes with goggles.

1.2 Disposal Instructions

We care about the environment! Here are some tips on how to dispose of our kit and its packaging responsibly:

- Electronic components and batteries must not be disposed of in household waste. Please take them to a collection point or recycling center.
- Our packaging is made of environmentally friendly materials. Recycle it in the appropriate bins.
- Separate plastic, paper, and other materials to facilitate recycling.

2 Basics

2.1 Soldering

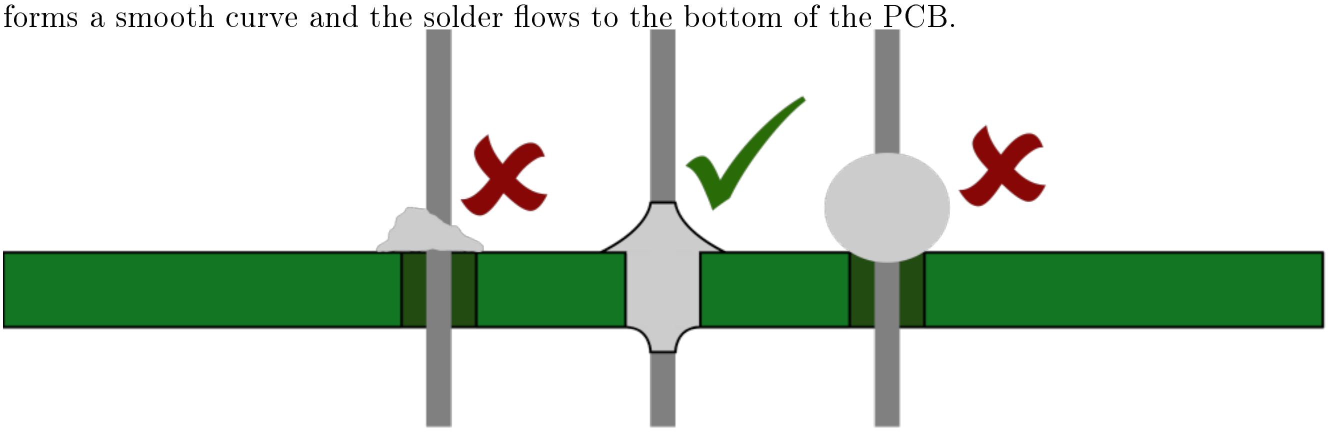
Our soldering exercises are designed to teach you the basics of soldering step by step. Don't worry, no one becomes an expert overnight! Just like in many things, *"practice makes perfect."*

You can find a helpful video about common soldering mistakes on our YouTube channel:
`https://youtu.be/CPXZM8r8xFw`

Here are some important points in brief:

- The ideal **temperature** for the soldering iron should be around **350°C**. It is recommended to use lead-free solder.
- The choice of soldering tip should match the size of the component. A **1.6 mm chisel tip** is ideal for most solder joints – it works for SMD components of size 0805 and wires up to 1.5 mm^2.
- To protect yourself from solder splashes or flying wires when cutting, it's recommended to wear **safety glasses**.
- Use a **heat-resistant surface** to avoid burn marks on your table.

The right amount of solder is crucial. Make sure the solder covers both the component and the PCB. The soldering iron must touch both the **component** and the **PCB** at the same time so that both are heated properly.
The following image shows typical solder joints: On the left, too little solder was used, while on the right, a ball forms – this happens when only the wire is heated. A perfect solder joint (middle) forms a smooth curve and the solder flows to the bottom of the PCB.



2.2 Resistors

Resistors are found in almost every circuit and are one of the three basic **passive components**, along with inductors and capacitors.
The job of a resistor is to **limit** the flow of current. The unit of resistance is the Ohm[1] (Ω). The relationship between current, voltage, and resistance is described by the well-known **Ohm's Law**:

$$U = R \cdot I$$
$$R = \frac{U}{I}$$

[1] Named after the German physicist Georg Simon Ohm, born in 1789

The **resistance value** of a component can be read differently depending on its type: For SMD[2] components, the value is printed as a code, while for through-hole resistors, it's shown by color bands.
SMD resistors usually have a three-digit number like 471. The first two digits give the value, and the last digit shows the number of zeros. So, 471 means 47 followed by one zero, or 470Ω. A code like 683 means 68 followed by three zeros, or 68000Ω (68kΩ).
For through-hole resistors, the value is read using color bands. The following image shows how this works:

Farbe	1. Stelle	2. Stelle	(3. Stelle)	Multiplikator	Toleranz
Silber				x0,01	±10%
Gold				x0,1	±5%
Schwarz	0	0	0	x1	
Braun	1	1	1	x10	±1%
Rot	2	2	2	x100	±2%
Orange	3	3	3	x1.000	
Gelb	4	4	4	x10.000	
Grün	5	5	5	x100.000	±0,5%
Blau	6	6	6	x1.000.000	±0,25%
Violett	7	7	7	x10.000.000	±0,1%
Grau	8	8	8	x100.000.000	±0,05%
Weiß	9	9	9	X1.000.000.000	

4	7	x100	±5%		= 4k7Ω
6	8	0	x1	±0,1%	= 680Ω

For SMD resistors, the resistance values can be identified based on their markings. The labeling works as follows: The first two digits indicate the value, and the third digit represents the number of zeros following the value. For example, a $220k\Omega$ resistor is marked as "224," and a 470Ω resistor is labeled as "471".

Don't worry about damaging resistors when soldering—they are very heat-resistant and unlikely to get damaged. Also, it doesn't matter which way you solder them because resistors have no polarity. For the perfectionists: It looks neater if all the color bands face the same direction.

2.2.1 Exercises

A resistor can limit the flow of current, which is often useful when you don't want the current to go over a certain value.
The following two examples show how to calculate the resistance using Ohm's law. Ohm's law is:

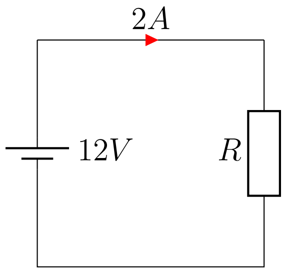
$$U = R \cdot I \quad \text{or} \quad R = \frac{U}{I}$$

Example 1:
Let's say we have a circuit with a voltage of $U = 12V$ and a current of $I = 2A$. To calculate the resistance, we put the values into the formula:

$$R = \frac{U}{I} = \frac{12V}{2A} = 6\,\Omega$$

So, the resistance in this circuit is $6\,\Omega$.

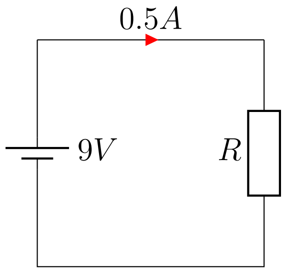
[2] Surface Mount Device



Example 2:
In another example, the voltage is $U = 9V$ and the current is $I = 0.5A$. The resistance is calculated as:

$$R = \frac{U}{I} = \frac{9V}{0.5A} = 18\,\Omega$$

So, the resistance in this circuit is $18\,\Omega$.



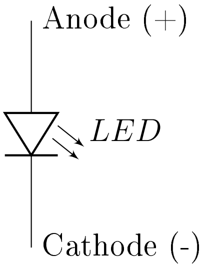
As you can see, it's easy to calculate the resistance when you know the voltage and current.

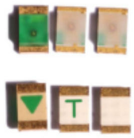
Calculation examples:

- A resistor with $R = 470\Omega$ is connected to a 5V power source. What is the maximum current? (Answer: $10.6mA$)
- A power supply of 24V should deliver a maximum current of $200mA$. What resistor do you need? (Answer: 120Ω)
- What is the voltage of a power supply if it provides a current of about 7mA through a 470Ω resistor? (Answer: $3.3V$)

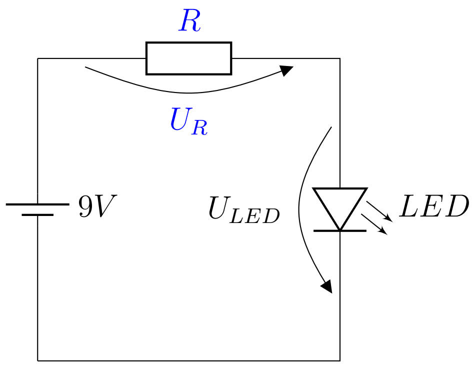
2.3 Soldering LEDs

A light-emitting diode (LED) is an electronic component that emits **light** when current flows through it.
The correct **polarity** of the LED is crucial. If it is soldered onto the board the wrong way, it will not light up and might even be damaged.
The LED has two terminals: the **anode** (positive, +) and the **cathode** (negative, -). You can distinguish between them by the length of the legs. An easy way to remember is: *"Minus is always shorter"* – the shorter leg marks the cathode. For SMD LEDs, the cathode is often marked with a colored dot on the top or a diode symbol on the bottom.





An LED should never be operated without a **current-limiting resistor**. This resistor limits the maximum current flowing through the LED and protects it from overloading and potential damage. Always check the LED's datasheet to determine the correct values.
Example for calculating a current-limiting resistor:



Let's say we have an LED that lights up at a voltage of $U_{LED} = 2V$ and requires a maximum current of $I_{LED} = 20mA$. The supply voltage is $U_S = 9V$.
Since $2V$ are dropped across the LED, the remaining $7V$ are left for the resistor, as calculated by Kirchhoff's voltage law:

$$U_R = U_S - U_{LED} = 9V - 2V = 7V$$

The current flowing through the LED is the same as the current through the resistor. So, we can calculate the resistor's value using Ohm's law:

$$R = \frac{U_R}{I_{LED}} = \frac{7V}{0.02A} = 350\,\Omega$$

Thus, the required resistor is $350\,\Omega$. Since resistors come in standard values (called E-series), you often need to round up. The closest available value would be $360\,\Omega$.
You can also find a video on how to calculate the current-limiting resistor for LEDs on our YouTube channel:
`https://youtu.be/p0dt-6F-EZg`

2.3.1 Exercises

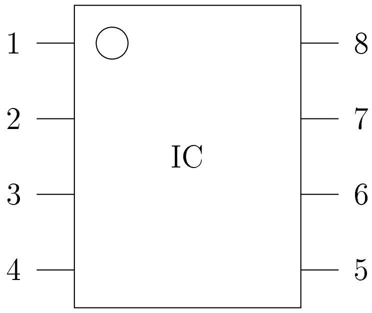
The answers are rounded to standard E-series resistor values!

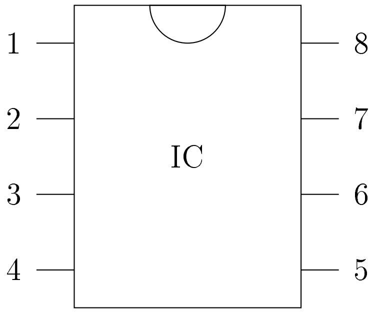
- You have a low-current LED running at $I = 2mA$ and requiring a voltage of $2.5V$. You want to power it with $5V$. What current-limiting resistor do you need? (Answer: 1300Ω)
- A blue LED requires $4V$ and a current of $17mA$. What resistor should you use to power it with $12V$? (Answer: 470Ω)

2.4 IC

Integrated Circuits (ICs) are compact components that come in many different types, ranging from simple logic gates to operational amplifiers and complex microcontrollers. Typically, they are small, black components with pins, either in through-hole or SMD technology.

An important thing to remember when handling ICs is the **polarity**, which is marked by a small dot or a notch on the case. This marking always indicates the position of **Pin 1**. When placing the IC on a circuit board, be sure to check the **correct orientation** of the polarity, as placing it incorrectly can damage the component.





You will also find a marking for Pin 1 on the circuit board, making it easier to correctly align the chip when soldering it.

2.5 Capacitors

Capacitors are passive electronic components used to store electrical charge. They usually consist of two conductive surfaces separated by an insulating material (dielectric).

Capacitance

The capacitance of a capacitor is measured in farads (F). It indicates how much electric charge a capacitor can store at a given voltage. In practice, common values range from picofarads (pF) to millifarads (mF).

$$C = \frac{Q}{U}$$

Where C is the capacitance, Q the stored charge in coulombs, and U the voltage in volts.

Maximum Voltage

Each capacitor has a rated maximum operating voltage. Exceeding this voltage can cause the dielectric to break down, damaging or destroying the capacitor. The allowable voltage is usually printed on the component (e.g., "16V").

Charging Curve of a Capacitor

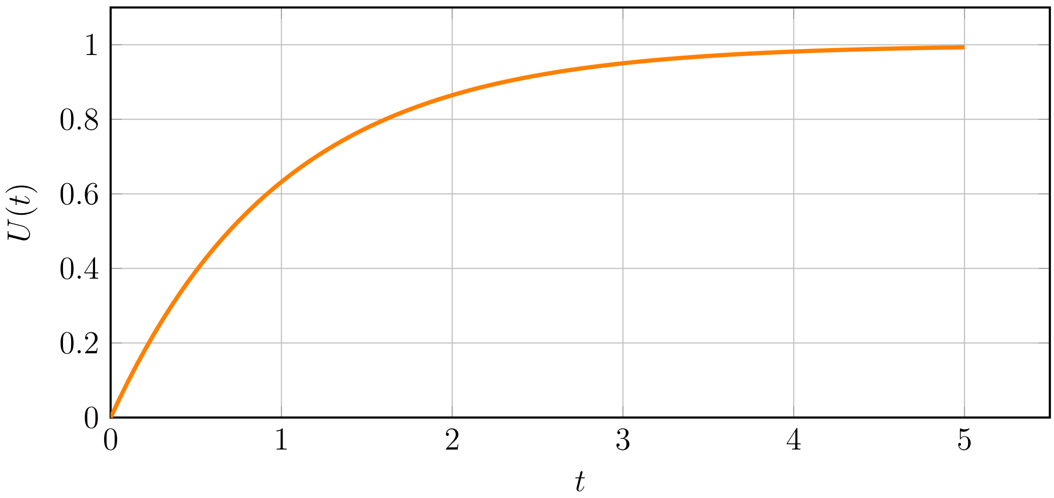
When charging a capacitor through a resistor, the voltage across the capacitor increases exponentially rather than instantly. The voltage $U(t)$ at time t is given by:

$$U(t) = U_0 \cdot \left(1 - e^{-t/RC}\right)$$

The time constant τ (tau) is the product of resistance R and capacitance C:

$$\tau = R \cdot C$$

After approximately five time constants (5τ), the capacitor is considered fully charged.



Polarity in Electrolytic Capacitors

Electrolytic capacitors have a fixed polarity that must be observed during installation. The negative terminal is usually marked with a stripe on the casing. Reversing the polarity can damage the capacitor or cause it to explode.

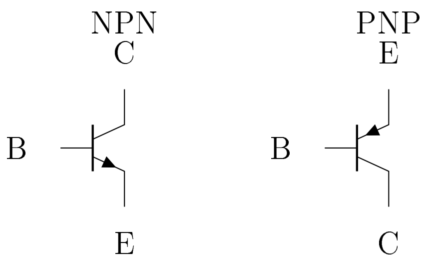
A capacitor stores electrical energy and smooths voltage fluctuations in circuits. Always respect polarity and voltage ratings!

2.6 Bipolar Transistors

Bipolar transistors are active electronic components used as switches or amplifiers. They consist of three layers of differently doped semiconductor material and have three terminals: base (B), collector (C), and emitter (E).

Structure: NPN and PNP

There are two main types: NPN and PNP. In an NPN transistor, current flows from collector to emitter when a small current flows from base to emitter. In a PNP transistor, the direction is reversed.



Gain (β)

The current gain β of a bipolar transistor indicates how much larger the collector current I_C is compared to the base current I_B:

$$\beta = \frac{I_C}{I_B}$$

Typical values for β range from 50 to 300. The exact value depends on the transistor type and is specified in the respective datasheet. Usually, a minimum and maximum value is given – β is therefore not exact, but rather a range.

Typical Circuits

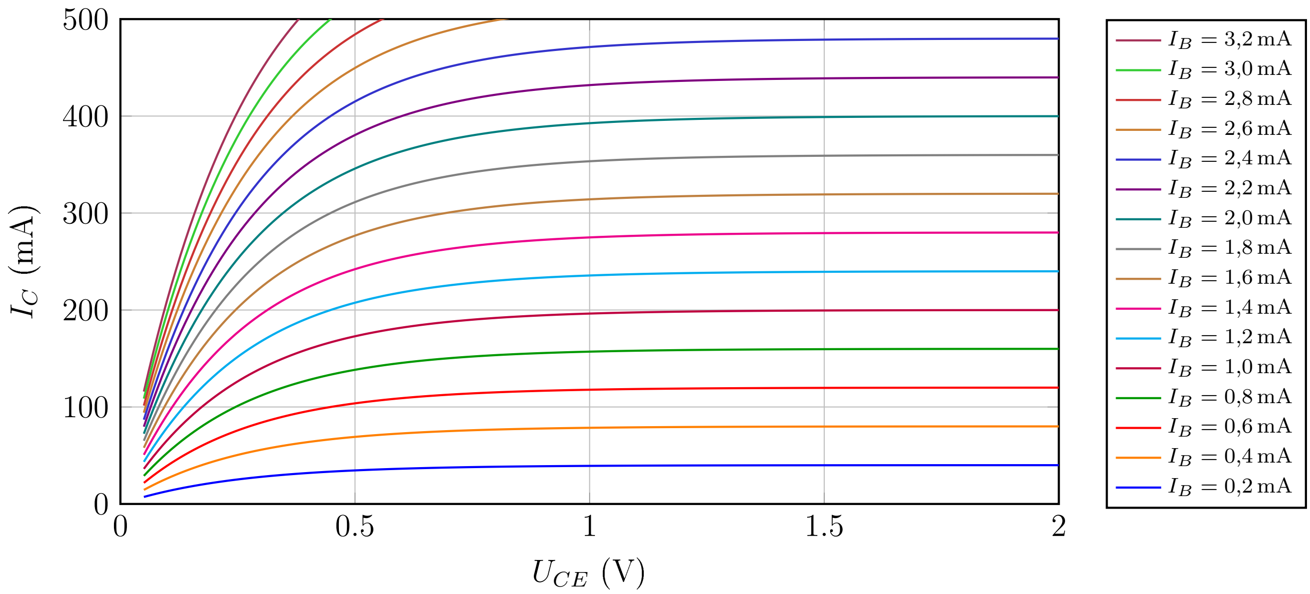
Bipolar transistors are commonly used in the following configurations:

- Common-collector (emitter follower)
- Common-emitter (standard amplifier)
- Common-base (rare, for special applications)

The common-emitter configuration is the most widely used amplifier setup.

Operating Point and Characteristic Curve

For the NPN transistor BC337, typical output characteristics at room temperature show the collector current I_C as a function of collector-emitter voltage U_{CE} for different base currents I_B:



> A bipolar transistor uses a small base current to control a much larger collector current, making it ideal for amplification and switching tasks.

Example Calculation for BC337

Let's consider a common-emitter circuit using a BC337 transistor. The supply voltage is $V_{CC} = 9\,\mathrm{V}$, the collector resistor is $R_C = 470\,\Omega$, the base resistor is $R_B = 100\,\mathrm{k\Omega}$, and the gain is $\beta = 200$. The base-emitter voltage is typically $U_{BE} = 0.7\,\mathrm{V}$.

- $V_{CC} = 9\,\mathrm{V}$
- $R_B = 100\,\mathrm{k\Omega}$
- $R_C = 470\,\Omega$
- $\beta = 200$
- $U_{BE} = 0.7\,\mathrm{V}$

1. Calculate the base current:

$$I_B = \frac{V_{CC} - U_{BE}}{R_B} = \frac{9\,\text{V} - 0.7\,\text{V}}{100\,000\,\Omega} = 83\,\mu\text{A}$$

2. Estimate the collector current:

$$I_C = \beta \cdot I_B = 200 \cdot 83\,\mu\text{A} = 16.6\,\text{mA}$$

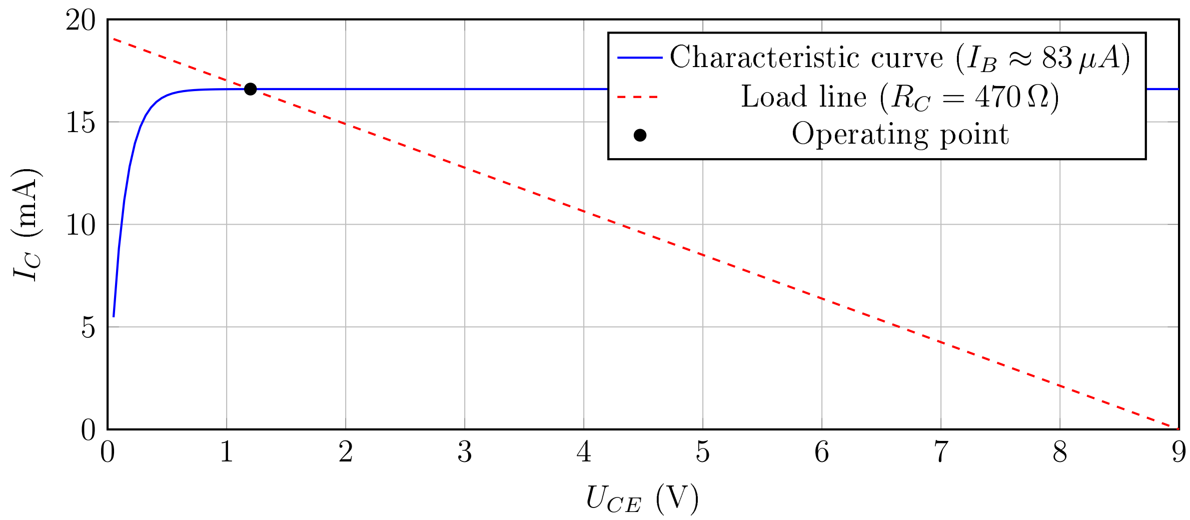
3. Determine the collector voltage:

$$U_{RC} = I_C \cdot R_C = 16.6\,\text{mA} \cdot 470\,\Omega = 7.8\,\text{V}$$

$$U_C = V_{CC} - U_{RC} = 9\,\text{V} - 7.8\,\text{V} = 1.2\,\text{V}$$

4. Operating point in the characteristic graph:
The following diagram shows the output characteristic curve of the transistor for $I_B = 83\,\mu A$, as well as the load line of the circuit. The operating point is at the intersection:

$$U_C \approx 1.2\,\text{V}, \quad I_C \approx 16.6\,\text{mA}$$



3 Solderexcercise police light

This kit is not a toy and is intended solely for learning how to solder. Use in road traffic is strictly prohibited!

3.1 Bill of Materials

The following components are required to build the kit. Please refer to the schematic and component placement plan to determine where each part belongs. It is recommended to assemble the kit in the order listed below:

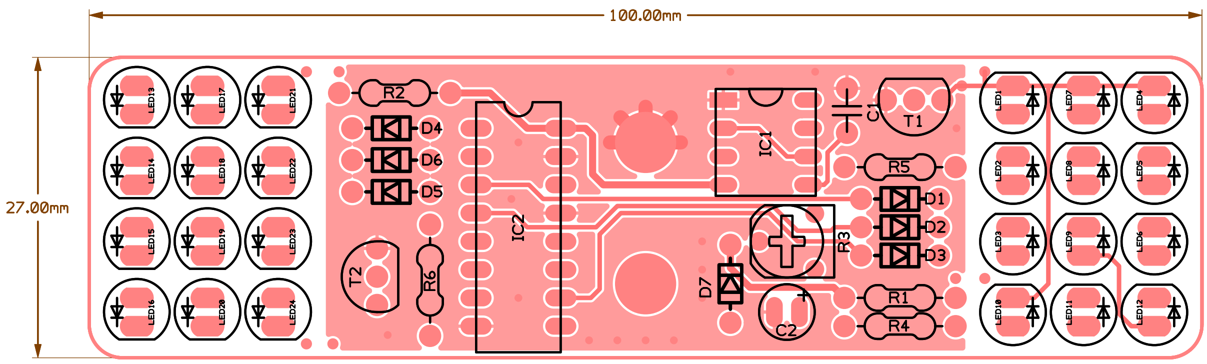
Table 1: Bill of Materials for the Police Light Kit

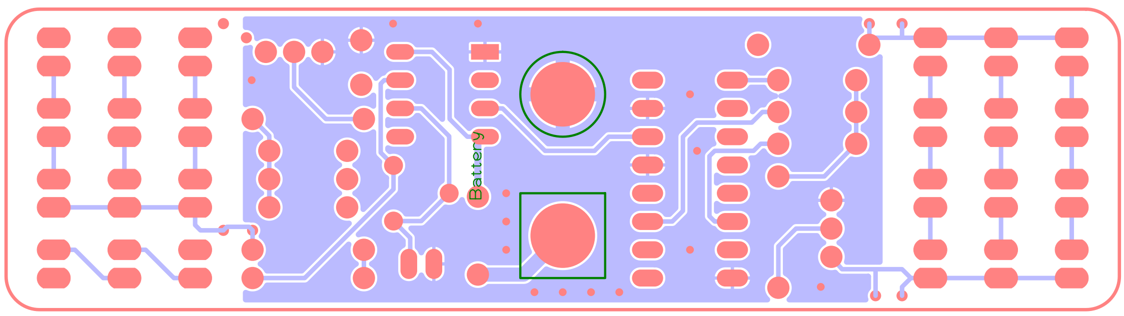
Name	Value	Quantity	Description
D1	1N4148	7	1N4148
D2			
D3			
D4			
D5			
D6			
D7			
R1	100	2	R_1/4W_generic
R2			
R3	100k	1	Poti_RKT6V-10K
R4	10k	3	R_1/4W_generic
R5			
R6			
C1	10n	1	C_Kerko_RM4_10nF
T1	BC337	2	BC337
T2			
LED1	blue	12	LED_5mm_blue
LED2			
LED3			
LED4			
LED5			
LED6			
LED7			
LED8			
LED9			
LED10			
LED11			
LED12			
LED13	red	12	LED_5mm_red
LED14			
LED15			
LED16			
LED17			
LED18			
LED19			

Continued on next page

Table 1: Bill of Materials for the Police Light Kit (continued)

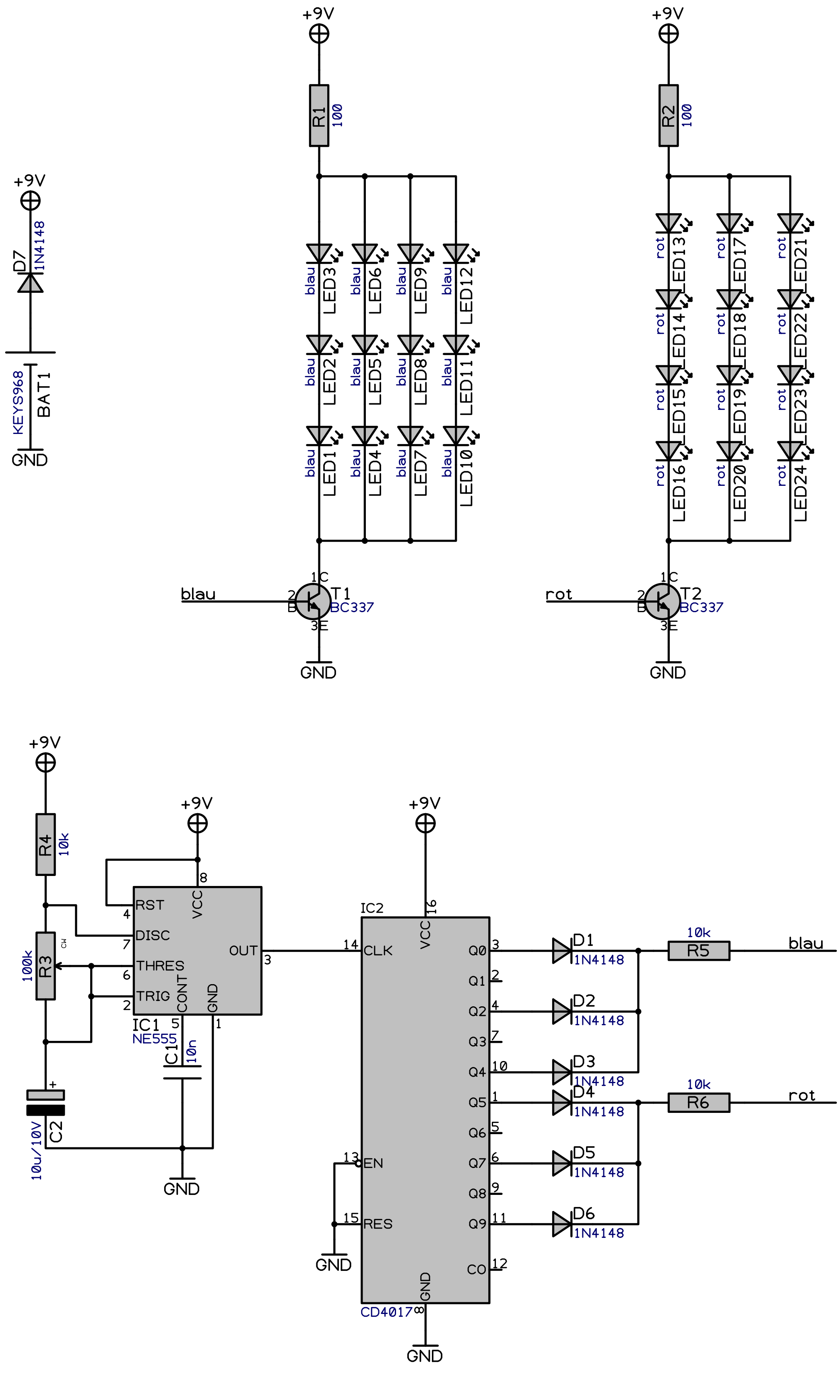
Name	Value	Quantity	Description
LED20			
LED21			
LED22			
LED23			
LED24			
C2	10u/10V	1	C_Elko_100u_10V_THT_RM2x5
IC1	NE555	1	NE555P
IC2	CD4017	1	CD4017
BAT1	KEYS968	1	Keystone 968





3.2 Schematic

The schematic for this kit can be found on the next page to ensure correct scaling when printed.



Autor	Kunde	Seitenname	
hajo	meinereiner	Schaltung	
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Elektronik Hannes Jochriem Oberfeldweg 12 83080 Oberaudorf Tel.: 08033 / 695 69 89 https://www.eHaJo.de	Titel, zusätzlicher Titel polizeilicht	Sachnummer 0815	Gespeichert 22.05.2025
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